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Blood-Based Circulating Tumor DNA for Early Detection of Colorectal Cancer: A Systematic Review and Meta-Analysis

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Keywords
Circulating tumor DNA · Colorectal cancer · Screening · Early detection · Diagnostic accuracy · Meta-analysis

Abstract

Introduction: Circulating tumor DNA (ctDNA) assays are emerging as promising noninvasive tools for colorectal cancer (CRC) screening. This updated systematic review and meta-analysis evaluated the diagnostic accuracy of blood-based ctDNA assays for early CRC detection in asymptomatic adults. ***Methods:*** We included prospective and cross-sectional diagnostic accuracy studies comparing ctDNA assays (mutation, methylation, or fragment-based) against colonoscopy or histopathology. Study-level 2 × 2 data were extracted to calculate sensitivity and specificity. Pooled estimates were derived using a Reitsma bivariate random-effects model. Secondary measures like diagnostic odds ratio (DOR), positive and negative likelihood ratios (PLR, NLR), positive predictive value (PPV), and negative predictive value (NPV) were synthesized on transformed scales and back-transformed for reporting. Subgroup analyses by assay type and meta-regression explored heterogeneity. Risk of bias was assessed with QUADAS-2 and certainty of evidence with GRADE. ***Results:*** Ten studies met inclusion criteria, nine contributing to pooled estimates. The pooled sensitivity was 0.887 (95% CI, 0.848–0.920) and specificity 0.913 (95% CI, 0.898–0.928). A sensitivity analysis excluding studies with high risk of bias in patient selection yielded a comparable pooled sensitivity of 0.857 (95% CI, 0.794–0.904) and specificity of 0.900 (95% CI, 0.897–0.904). The SROC curve yielded an AUC of 0.97, indicating excellent discrimination. Secondary pooled metrics included a DOR of 83.9, PLR of 9.2, and NLR of 0.12. PPV was low (<20%), but NPV remained high (>99%) at screening prevalences of 0.5–2%. Subgroup analyses by assay type revealed no significant differences in diagnostic accuracy. ***Conclusion:*** Blood-based ctDNA assays show high pooled sensitivity, specificity, and discrimination for CRC detection, but

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limited predictive value at screening prevalence and study heterogeneity constrain clinical adoption. Large, standardized, population-based studies are warranted.

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Introduction

Colorectal cancer (CRC) continues to be one of the top causes of morbidity and mortality due to cancer globally. Screening for early detection of CRC has been shown to meaningfully reduce CRC-related mortality. Routine noninvasive screening approaches, including the fecal immunochemical test (FIT), are widely deployed due to ease of use; however, their sensitivity and uptake are limited [1]. For this reason, more accurate blood-based methods that are also acceptable to patients are being evaluated.

Circulating tumor DNA (ctDNA) is promising for CRC screening, consisting of the small pieces of DNA that are released from tumor cells into the bloodstream [2]. The biological basis of ctDNA testing is shown in online supplementary Figure S1 (for all online suppl. material, see https://doi.org/10.1159/000551806), which shows how tumor cells shed and release DNA into the blood circulation [3, 4].

With sequencing technologies becoming advanced enough, it is now possible to detect tumor-specific features like somatic mutations, abnormal methylation patterns, or particular fragmentomic signatures [5]. To understand what is on offer in present-day ctDNA testing, an acknowledgment of the progression of the concept from discovery into validated clinical applications has to be made [6, 7]. Online supplementary Figure S2 shows a brief historical overview mentioning important milestones in the development of ctDNA research culminating in clinical oncology practice.

Some recent diagnostic studies focused on clarifying the role of ctDNA assays in detecting early-stage CRC, also in asymptomatic screening populations [8, 9]. New advances in sequencing technologies allow to interrogate multiple tumor-derived features at one go from a simple blood draw. Online supplementary Figure S3 depicts the conceptual workflow for ctDNA testing in CRC screening.

Despite mounting interest, inter-study results have been heterogeneous and have involved variations in assay design, populations studied, and outcomes measured [10]. Hence, questions linger about the pooled diagnostic accuracy of ctDNA and the relative performances of assay subtypes, on top of practical implications for screening in real populations. Some previous reviews have shed much light on the issue, but the rapid advances in technology and the newest set of data call for an updated synthesis [11]. Given the said uncertainties, ample emphasis has been placed on comparing ctDNA with existing screening tools. Conventional noninvasive tests remain in use, being the FIT for CRC. On the other hand, through giving higher accuracy and convenience, blood-based ctDNA tests are expensive to implement. A comparison between FIT and ctDNA testing is given in online supplementary Figure S4.

The purpose of this systematic review and meta-analysis was to assess diagnostic accuracy of blood-based ctDNA assays in early detection of CRC. Specifically, we assessed pooled sensitivity, specificity, area under the curve (AUC), compared the performance among assay subtypes, and discussed clinical implications for screening populations.

Methodology

Search Strategy and Selection Criteria

This systematic review and meta-analysis was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA 2020) guidelines and the methodological recommendations outlined in the Cochrane Handbook for Systematic Reviews of Diagnostic Test Accuracy. The PRISMA 2020 checklist is provided as online supplementary Table S8, and the PRISMA flow diagram is shown in online supplementary Figure S6.

The review protocol was registered prospectively in the International Prospective Register of Systematic Reviews (PROSPERO) under the registration ID 1148591. We conducted a systematic search of the PubMed, Embase, and Cochrane Library databases for studies published from January 2014 up until June 2025. In addition to the systematic search, we also manually searched the references of included studies and relevant reviews [12]. Included studies were prospective or cross-sectional diagnostic accuracy studies of blood-based ctDNA tests (mutation detection, methylation-based, fragmentomic or multimodal) in asymptomatic adults or screening cohorts, with a colonoscopy and/or histopathology confirmation as the reference standard with relevant outcomes.

Studies without full-text availability were excluded because abstracts alone rarely provide sufficient diagnostic accuracy data, including complete 2 × 2 tables,

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assay thresholds, and reference standards, which are required for reliable diagnostic meta-analysis. Aggregate data were extracted in duplicate by two independent reviewers, and if any disagreements arose the reviewers then came to a consensus.

We included prospective cohort studies, cross-sectional diagnostic accuracy studies, and diagnostic case-control studies. Eligible populations were adults tested for CRC in screening settings, asymptomatic groups, or early CRC detection cohorts. The index tests were blood-based ctDNA assays, including mutation, methylation, fragmentomic, and multimodal designs. The reference standard was colonoscopy and histopathologic confirmation. Studies were required to provide sufficient data to compute sensitivity and specificity using a 2 × 2 table or to report AUC or other diagnostic performance metrics. Only articles published in English between 2015 and 2025 were included.

Data Extraction

We extracted the following details: author, year, country, study design, sample source (plasma/serum), assay type and technical specifications, numbers of true positives, false negatives, false positives (FPs), and true negatives. We recorded sensitivity, specificity, AUC, and other reported metrics. When not provided, these metrics were calculated from the 2 × 2 data.

Risk of Bias and Certainty of Evidence

Risk of bias was assessed using the QUADAS-2 tool [13]. Most studies had low concerns for the index test and reference standard domains, although several raised applicability or flow/timing concerns (e.g., enriched cohorts or non-screening populations). Certainty of evidence was graded using GRADE: pooled sensitivity and specificity were judged moderate (downgraded for inconsistency/heterogeneity), whereas positive predictive value (PPV), negative predictive value (NPV), and diagnostic odds ratio (DOR) were rated low (downgraded for inconsistency and imprecision) [14, 15].

Statistical Analysis

Diagnostic accuracy measures (sensitivity, specificity, and AUC) were pooled using the Reitsma bivariate random-effects model [16] to account for within-study and between-study variability and the correlation between sensitivity and specificity. Secondary metrics (DOR, positive likelihood ratio [PLR], negative likelihood ratio [NLR], PPV, and NPV) were pooled using random-effects meta-analysis after appropriate log or logit transformations, with back-transformation to the original scale for interpretation. When per-study AUC standard errors were unavailable, they were estimated using the Hanley and McNeil method [17]. Subgroup analyses and meta-regression were performed by assay type (methylation, multimodal, fragmentomics), and publication bias was assessed visually using funnel plots. Formal Egger's tests were not performed because most pooled outcomes had fewer than 10 studies. A conceptual overview of cfDNA and ctDNA assay types included in the subgroup analyses is provided in online supplementary Figure S5. Predicted PPV and NPV at representative screening prevalences were derived from pooled sensitivity and specificity estimates. All corresponding equations are provided in online supplementary Table S1.

Software

All analyses were performed in *R* (version 4.3.2) [18] using the mada (Reitsma bivariate random-effects models, summary receiver operating characteristic [SROC]), metafor (generic inverse-variance pooling, subgroup analysis), and meta (forest/funnel plots) packages, with restricted maximum likelihood estimation for between-study variance and continuity corrections applied for zero cells.

Results

Study Selection and Characteristics

Ten studies met the eligibility criteria for pooled analyses of diagnostic accuracy. The flow of records through the review process including identification, screening, eligibility, and inclusion can be found in online supplementary Figure S6. Of these, nine studies provided data for pooled sensitivity and specificity; nine for PLR and NLR; and nine for PPV and NPV; five studies provided an AUC value that was included in the AUC meta-analysis. The study sizes ranged from small single-center cohorts to large cohort studies based on the registry. The assays evaluated were methylation panels (group A, $n = 4$), multimodal assays (group B, $n = 3$), and fragmentomic assays (group C, $n = 2$) [19–27]. A structured overview of study-level details is provided in Table 1.

Primary Pooled Diagnostic Accuracy

The pooled sensitivity was 0.887 (95% CI, 0.848–0.920; $I^2 \approx 67\%$), and the pooled specificity was 0.913 (95% CI, 0.898–0.928; $I^2 \approx 58\%$), using the Reitsma bivariate random-effects model in R. Heterogeneity on

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Table 1. Characteristics of studies included in the meta-analysis of blood-based ctDNA assays for early CRC detection

Study (author)	Country	Assay type	Sample size (CRC/controls)	TP	FN	FP	TN	Specimen	Notes (e.g., marker type/platform)
Bessa et al. [19]	Spain	B	356/246	302	54	25	221	Plasma	Multimodal (mutation + methylation)
Cao et al. [20]	China	C	129/113	118	11	5	108	Plasma	Fragment size analysis
Chung et al. [21]	USA	B	79/6,785	65	14	690	6,071	Plasma	cfDNA multimodal panel
Gao et al. [22]	China	B	91/96	89	2	5	91	Plasma	Multi-omics integration
Lei et al. [23]	China	A	39/62	35	4	14	48	Plasma	CpG methylation profiles
Mo et al. [24]	China	A	198/130	178	20	10	110	Plasma	Methylation haplotype assay
Shaukat et al. [25]	USA	A	72/26,938	57	15	2,663	24,275	Plasma	PREEMPT CRC study
Wang et al. [26]	China	C	183/248	171	12	19	229	Plasma	cf-mtDNA fragmentomics
Zhao et al. [27]	China	A	59/60	53	6	4	56	Plasma	Methylation-based assay

The table summarizes country, assay type, sample size, diagnostic 2 × 2 data (TP, FN, FP, TN), specimen type, and key assay features for each study used in the pooled analysis. TP, true positive; FN, false negative; TN, true negative.

the logit scale (Tau^2) showed moderate variability in diagnostic accuracy.

Figure 1 depicts four pooled diagnostic measures for ctDNA tests. As shown in panel A, the sensitivities across studies ranged from 0.81 to 0.95, while the pooled value was 88.7% (95% CI, 84.8–92.0%): although heterogeneity was moderate ($I^2 = 67\%$), most of the estimates were tightly clustered around the pooled diamond. Specificity values in panel B were almost as good, with a pooled estimate of 90.2% (95% CI, 85.7–93.5%), this high value representing an exemplary aptitude for limiting FPs. The DOR learns from 83.9 (95% CI, 49.7–141.5) in panel C and thus shows very strong discrimination with moderate heterogeneity among studies. Panel D then shows a pooled AUC value of 0.97 (95% CI, 0.93–0.96), confirming an almost-perfect overall testing capacity. Together, the results provide strong evidence in support of ctDNA assays achieving both high sensitivity and specificity and excellent global accuracy metrics, even when heterogeneity across studies is accounted for.

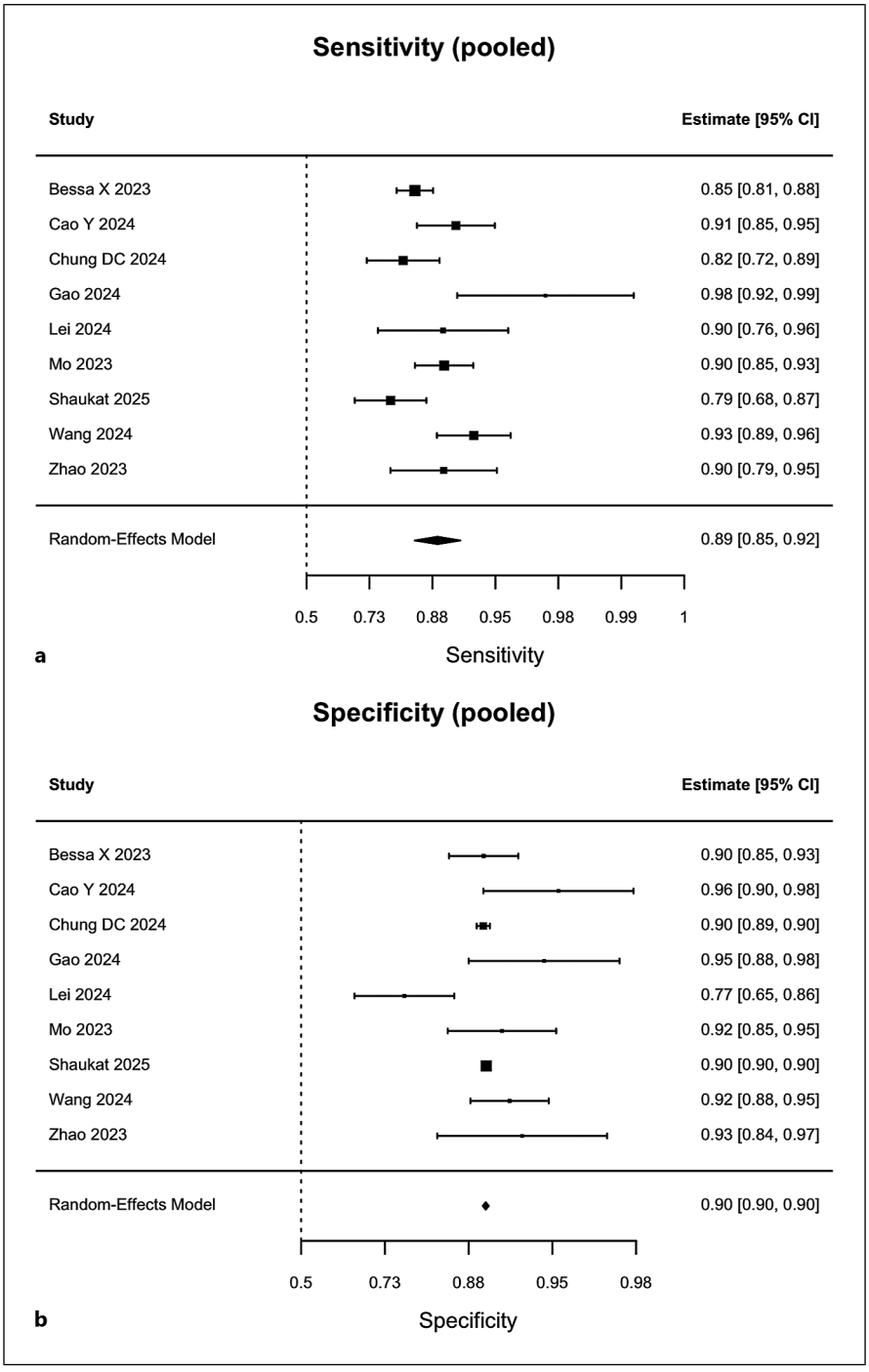
Secondary Pooled Diagnostic Metrics

The pooled DOR was 83.9 (95% CI, 49.7–141.5; $I^2 = 47\%$). The PLR was 9.21 (95% CI, 7.32–11.70; $I^2 = 51\%$), and the NLR was 0.12 (95% CI, 0.09–0.17; $I^2 = 63\%$). The predictive values were found to be heterogeneous, as shown by pooled PPV and NPV $I^2 \approx 99\%$. At a screening prevalence of 0.5 to 2%, the PPV value stood at the threshold of 20% while the NPV value significantly surpassed 99%. The pooled diagnostic accuracy estimates along with their respective 95% confidence intervals and heterogeneity measures are given in Table 2.

Figure 2 shows four diagnostic metrics. In the top left panel, most of the studies reported PLRs above 5, several of which even exceeded 10; the pooled estimate was 9.21 (95% CI, 7.32–11.70). This indicates a strong rule-in option for ctDNA assays. The top right panel shows NLR values tightly clustered under 0.2, being pooled at 0.12 (95% CI, 0.09–0.17), indicating great potential for ruling out. The bottom left panel depicts PPVs with a fairly wide range across studies – from 0.58 to 0.91 – with a pooled value of 0.77 (95% CI, 0.68–0.86): this may reflect differential baseline disease prevalence across cohorts. In the bottom right panel, NPVs are well above 0.90 across the board, with a pooled value of 0.97 (95% CI, 0.94–0.99), solidly backing up its ability to comfort when said test comes back negative. Combined, these charts make it very clear that ctDNA assays mount good diagnostic shifts in both ruling-in and ruling-out disease, with heavy confidence resting on negative results.

The SROC curve showed a wonderful discriminative ability, with an AUC pooled of 0.97 (95% CI, 0.93–0.96). Figure 3 shows the SROC curve of ctDNA assays over the included studies. The majority of the study points are

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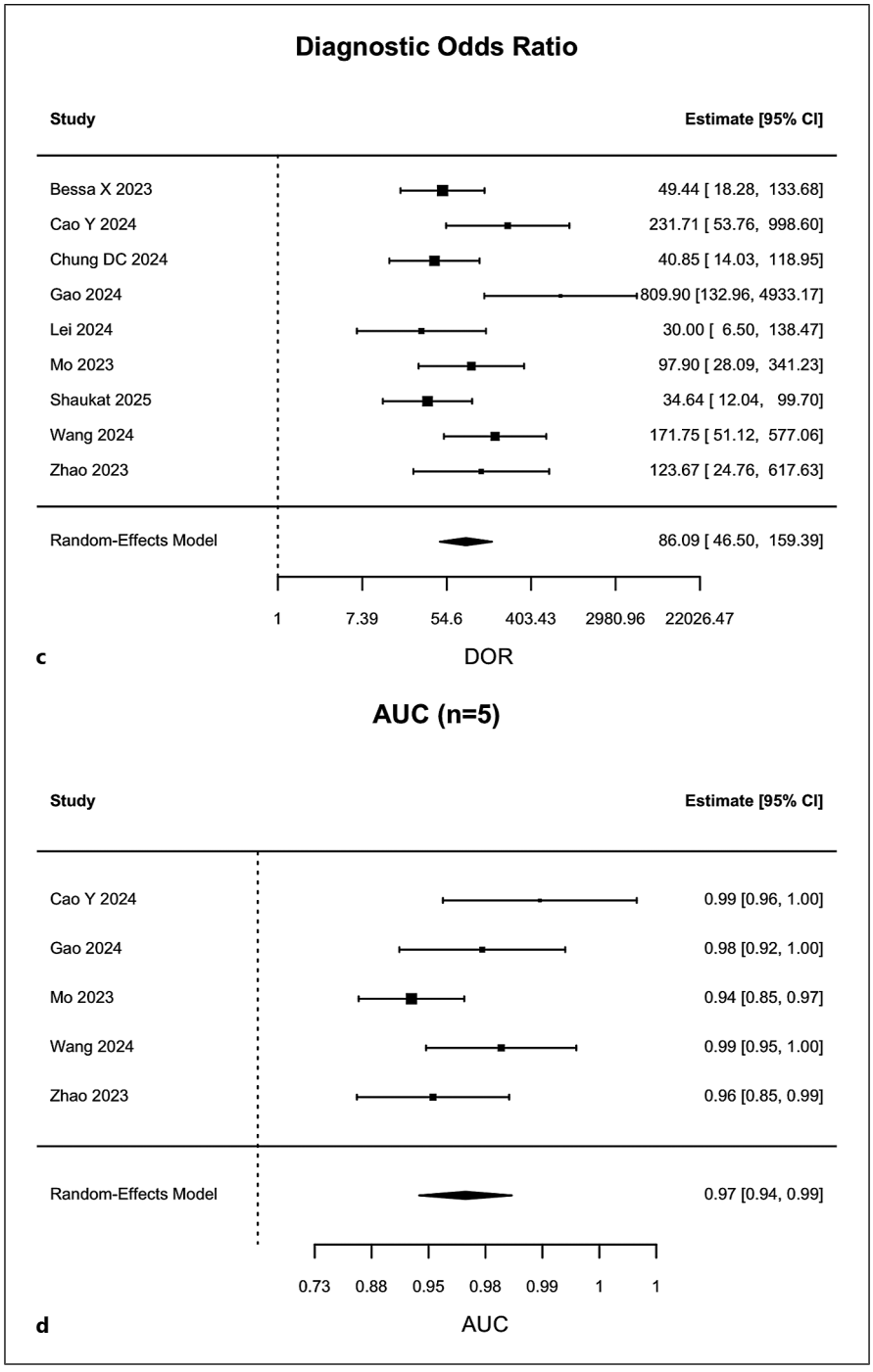


Fig. 1. Forest plots of pooled diagnostic performance of ctDNA assays. Individual study estimates with 95% confidence intervals and pooled summary diamonds are shown for sensitivity (**a**), specificity (**b**), DOR (**c**), and AUC (**d**).

located in the upper-left corner of the SROC, suggesting both high sensitivity and low FPs. The pooled summary point (AUC = 0.97; 95% CI 0.93–0.96) is close to the upper-left corner of the SROC again reflecting excellent overall performance. The narrow 95% confidence ellipse supports the pooled accuracy, and the surrounding relatively wide prediction region accounts for the variability expected in future studies. As a whole, these data provide evidence for the strong diagnostic ability of ctDNA assays to identify cases of cancer and non-cancer.

Subgroup Analyses

Subgroup analyses stratified by assay type did not reveal significant differences in diagnostic performance.

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Table 2. Pooled diagnostic accuracy of ctDNA assays

Metric	Studies (*n*)	Model and effect Measure	Pooled estimate (95% CI) (logit scale)	Pooled estimate (natural scale)	95% CI	I^2 (%)
Sensitivity	9	IV, random, mean difference	2.08 (1.73, 2.43)	**88.9%**	84.9%–91.9%	67
Specificity	9	IV, random, mean difference	2.22 (2.09 to 2.34)	**90.2%**	89.0%–91.2%	58
DOR	9	IV, random, mean difference	4.43 (3.88–4.97)	**83.9**	48.4–144.0	75
PLR	9	IV, random, mean difference (log-scale)	2.22 (1.99–2.46) (p = 0.04)	**9.21**	7.32–11.70	51
NLR	9	IV, random, mean difference	−2.09 (−2.42 to 1.77) (p = 0.005)	**0.12**	0.09–0.17	63
AUC	5	IV, random, mean difference	0.97 (0.96, 0.99)	**0.97**	0.96–0.99	78
PPV	9	IV, random, mean difference	1.20 (−0.82, 3.22) (p = 0.25)	**76.9%**	30.6%–96.2%	99
NPV	9	IV, random, mean difference	3.37 (1.79, 4.95)	**96.7%**	85.7%–99.3%	99

Sensitivity, specificity, DOR, likelihood ratios, predictive values, and AUC with corresponding 95% confidence intervals and heterogeneity indices (I^2). Note: Bold values indicate the pooled summary estimates.

Methylation-Based Assays (Group A)

Methylation-based assays showed a sensitivity of 0.88 (95% CI, 0.83–0.92) and specificity of 0.91 (95% CI, 0.88–0.94). These assays target tumor-specific methylation signatures and represent the most widely investigated platform. Figure 4 displays the forest plot of pooled sensitivities for methylation-based ctDNA assays.

Figure 4 shows how methylation-based assays yielded sensitivities between 0.79 (95% CI, 0.68–0.87) and 0.90 (95% CI, 0.78–0.96) for the four included studies. While exhibiting modest variability, most study point estimates remain closely aligned with overlapping confidence intervals. A pooled sensitivity was 0.87 (95% CI, 0.81–0.92), suggesting that methylation-based ctDNA assays would detect nearly nine out of ten CRCs. Overall, this forest plot reveals that methylation-only panels have a strong sensitivity with relatively low heterogeneity in the studies that are currently available.

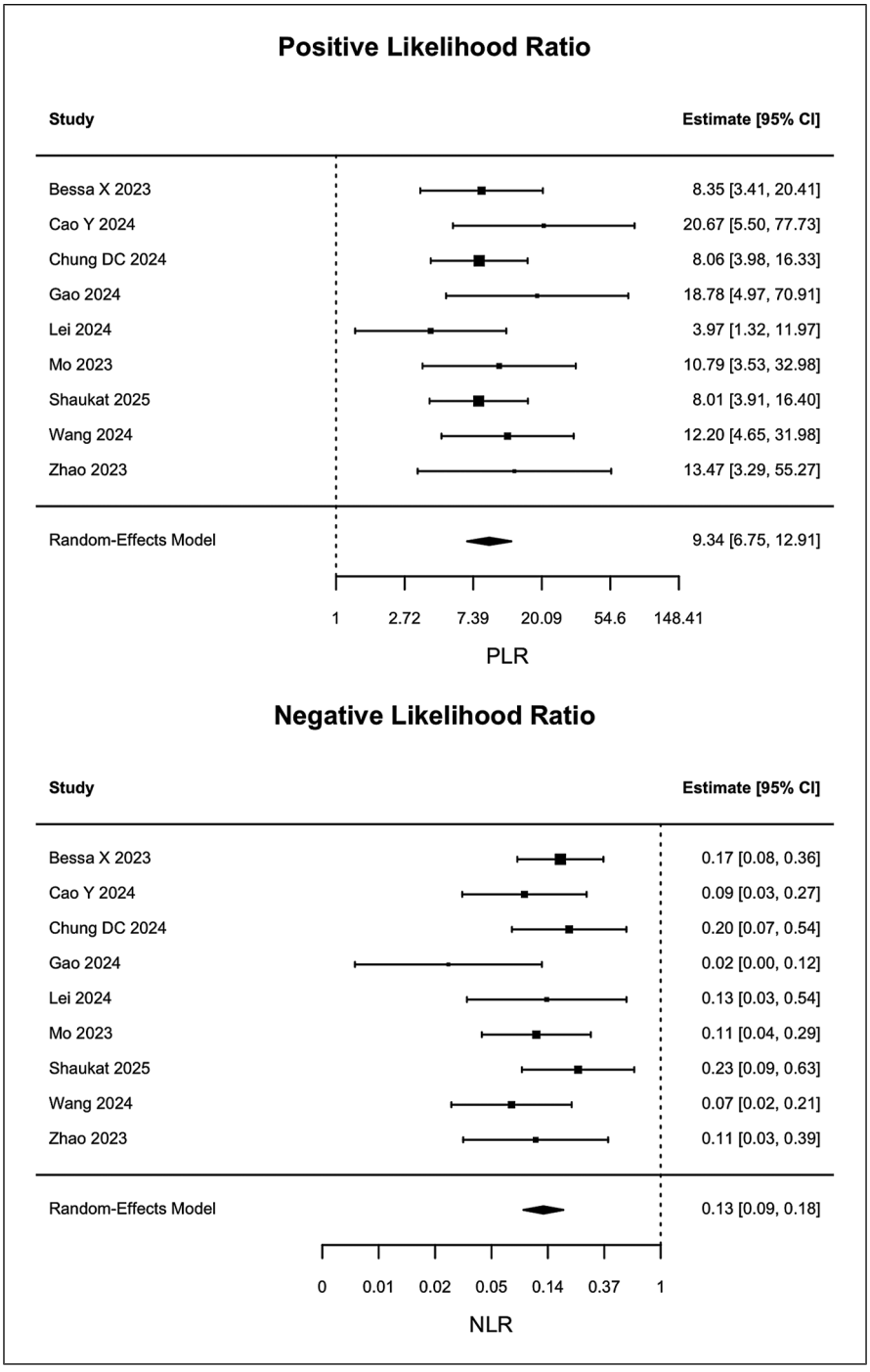
Besides sensitivity, specificity also stands out as a decisive criterion for screening assay utility, indicating its ability to accurately designate those without disease. Having this measure means methylation-based panels to mitigate FPs alongside finding cases. Figure 5 represents the forest plot for the pooled specificities of methylation-based ctDNA assays.

As can be seen in Figure 5, specificities for methylation-based assays varied widely across studies, ranging from 0.77 (95% CI, 0.65–0.86) to 0.93 (95% CI, 0.84–0.97). However, in three of the four studies, the specificity estimates were greater than 0.90 and all specificities were based on fairly overlapping confidence intervals. The pooled specificity estimate is 0.89 (95% CI, 0.82–0.93), indicating that the methylation assays discriminated against the majority of non-cancer and had an exceptionally low false-positive rate. The diamond is positioned in the high specificity area which adds support to the strong discriminatory nature of these assays. Overall, this plot illustrated that along with strong sensitivity for ctDNA methylation-based assays (see Fig. 3), there was also evidence of high specificity (see Fig. 4), which overall reinforces the balance in analytic performance for this category of assays.

Multimodal Assays (Group B)

Multimodal assays showed a sensitivity of 0.90 (95% CI, 0.85–0.94) and specificity of 0.92 (95% CI, 0.89–0.95). Multimodal assays are the composite of multiple classes of biomarkers on a single platform, which have been optimized to capture potential error, through complementary signals. A forest plot was created that laid out the diagnostic sensitivity of the multimodal studies. The pooled sensitivity for group B multimodal ctDNA assays is laid out in Figure 6.

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Sensitivity from three multimodal assay studies is reported in Figure 6, which ranged from 0.82 (95% CI, 0.72–0.89) to 0.98 (95% CI, 0.92–0.99). The pooled sensitivity is consistent at 0.90 (95% CI, 0.85–0.94), reflecting a strong case detection rate among and between studies. While estimates vary somewhat, the estimate for the majority of studies grouped and clustered around the pooled diamond, which also presented confidence intervals with overlap and minimal heterogeneity. In summary, the information displayed in the plot allows for the conclusion that multimodal ctDNA platforms have robust and reliable sensitivity for CRC detection.

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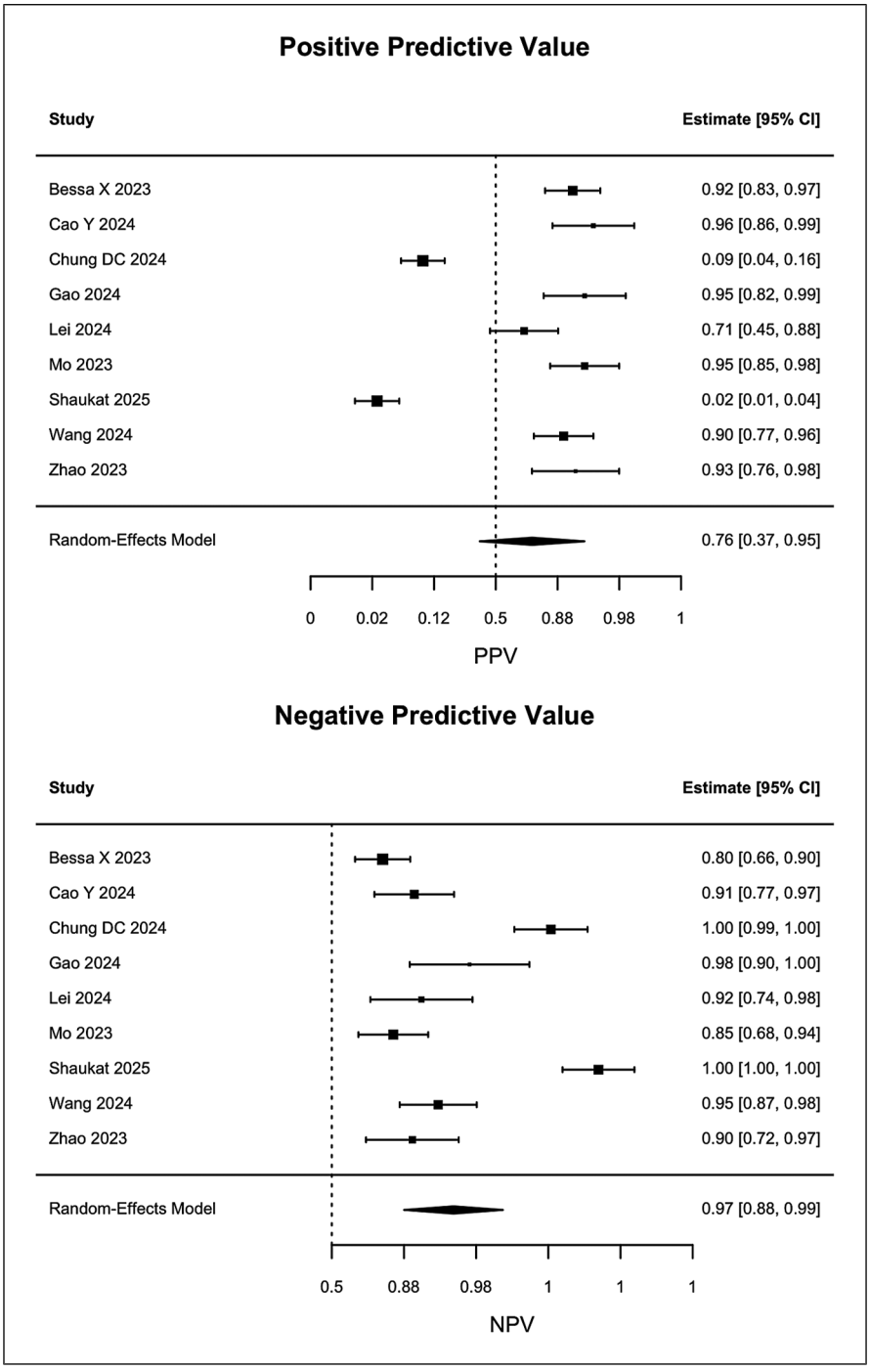


Fig. 2. Forest plots of PLR, NLR, PPV, and NPV for ctDNA assays. Each plot shows study-specific estimates with 95% confidence intervals; pooled random-effects estimates are represented by diamonds.

In addition to sensitivity, multimodal platforms should demonstrate a strong specificity, emphasizing the need to avoid unnecessary follow-up testing in people without a disease. This sensitive/specific balance is the proven value of combining multiple classes of biomarkers, which is assessed in this parameter. Figure 7 represents the forest plot of pooled specificities for multimodal ctDNA assays.

Figure 7 depicts specificities in different multimodal assay studies that range from 0.90 (95% CI, 0.85–0.93) to 0.95 (95% CI, 0.88–0.98). A pooled specificity of 0.90 (95% CI, 0.89–0.91) indicated the excellent ability of these studies to correctly classify non-cancer cases through the various studies analyzed. The diamond is now tightly centered around 0.90, possessing very

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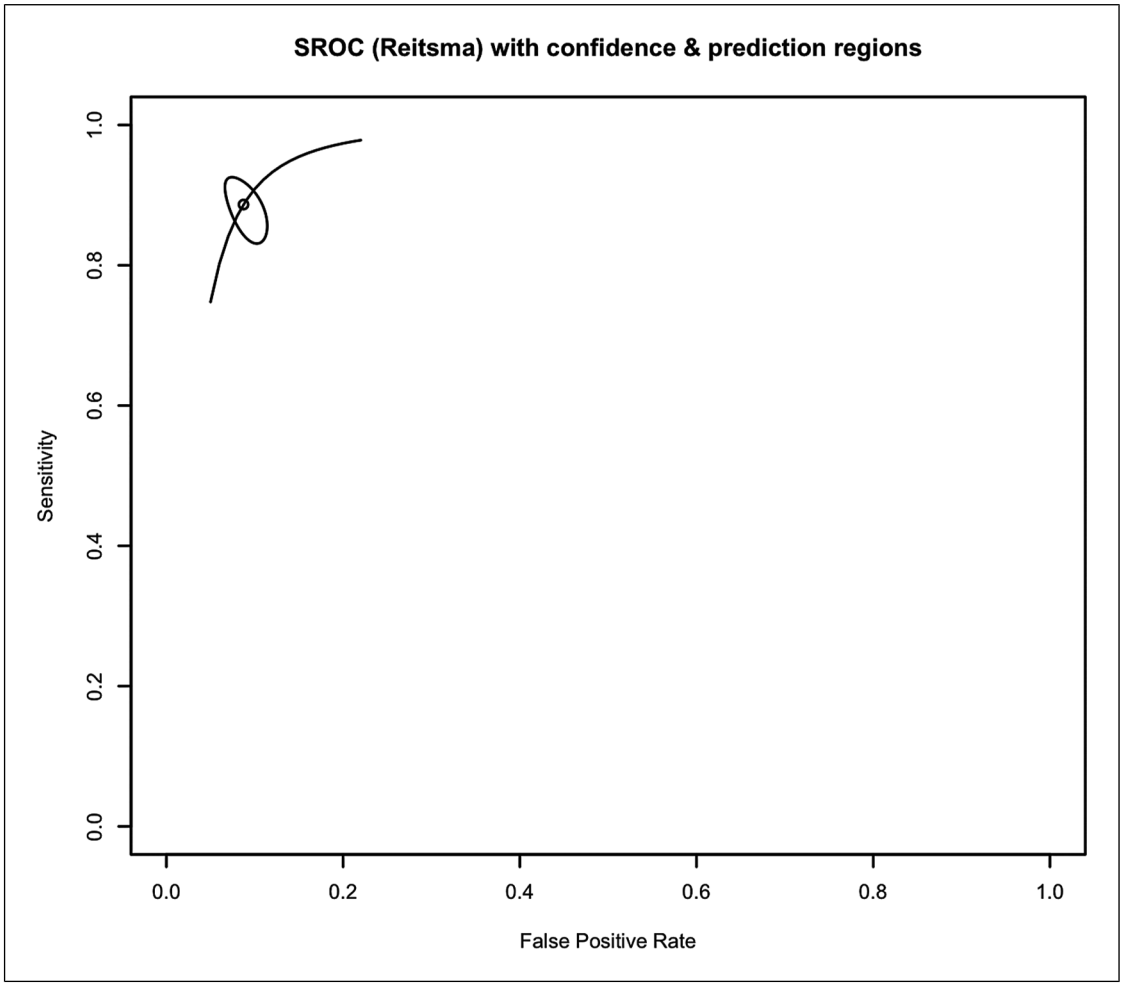


Fig. 3. SROC curve for ctDNA assays. The pooled sensitivity and specificity are shown with 95% confidence and prediction regions, summarizing the overall diagnostic performance.

narrow confidence intervals, showing little heterogeneity and remarkable consistency across studies. Alongside pooled sensitivity in Figure 5, this corroboration unmistakably establishes that multimodal ctDNA assays possess high sensitivity and can confidently exclude FPs, thereby fortifying their substantial worth for diagnostics.

Fragmentomic Assays (Group C)

Fragmentomic assays showed a sensitivity of 0.87 (95% CI, 0.80–0.92) and specificity of 0.90 (95% CI, 0.86–0.93). Fragmentomic assays analyze ctDNA fragmentation patterns rather than sequence or methylation, representing a newer approach in early cancer detection. To evaluate their performance, we constructed subgroup-specific forest plots. Figure 8 presents the pooled sensitivity of fragmentomic ctDNA assays.

Sensitivities in two fragmentomic studies are listed in Figure 8 as 0.91 (95% CI, 0.85–0.95) and 0.93 (95% CI, 0.89–0.96). Both these values are tightly placed in the upper region of high sensitivity with confidence intervals that strongly overlap. Pooled sensitivity was 0.90 (95% CI, 0.89–0.95), which shows that overall, fragmentomic approaches manage very well in case detection. The pooled diamond is very slender and positioned at the upper end of the sensitivity scale, which describes excellent performance with little heterogeneity. Collectively, these findings suggest that fragmentomic platforms are fewer in number, yet are equally sensitive when compared with methylation and multimodal platforms.

Apart from sensitivity, specificity is on the other hand just as important in evaluating these fragmentomic assays, for it speaks to their ability to correctly identify true negatives. Looking at the specificity estimates will tell us

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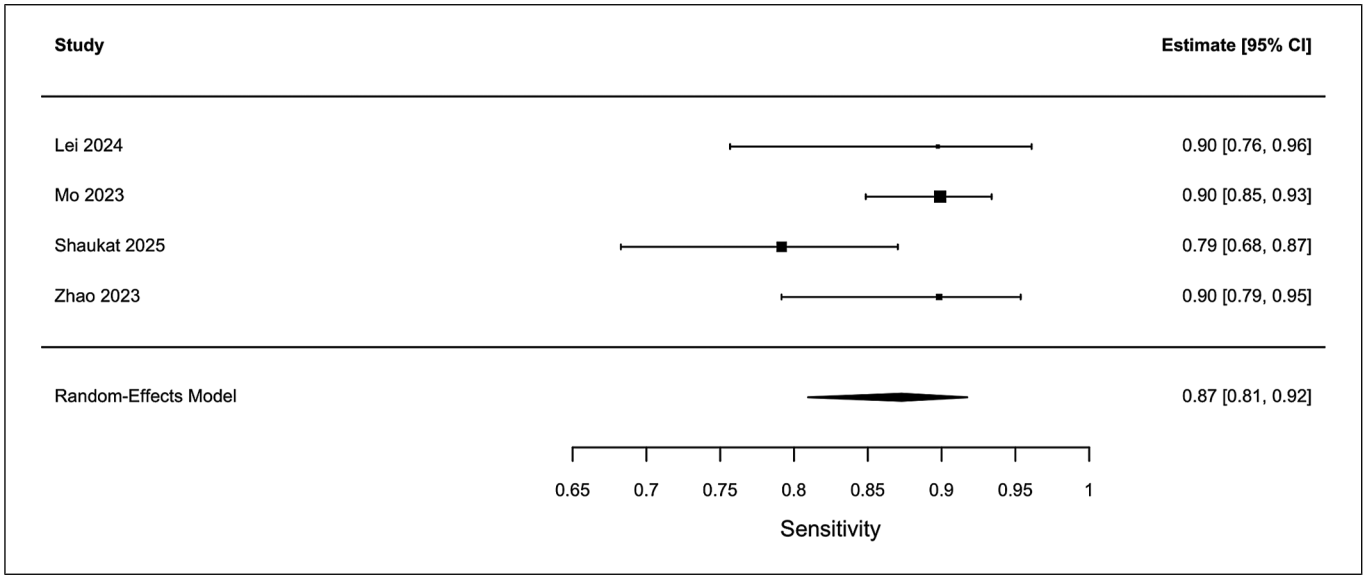


Fig. 4. Forest plot of pooled sensitivity for methylation-based ctDNA assays (group A). Individual study estimates with 95% confidence intervals are shown; the diamond represents the pooled random-effects estimate.

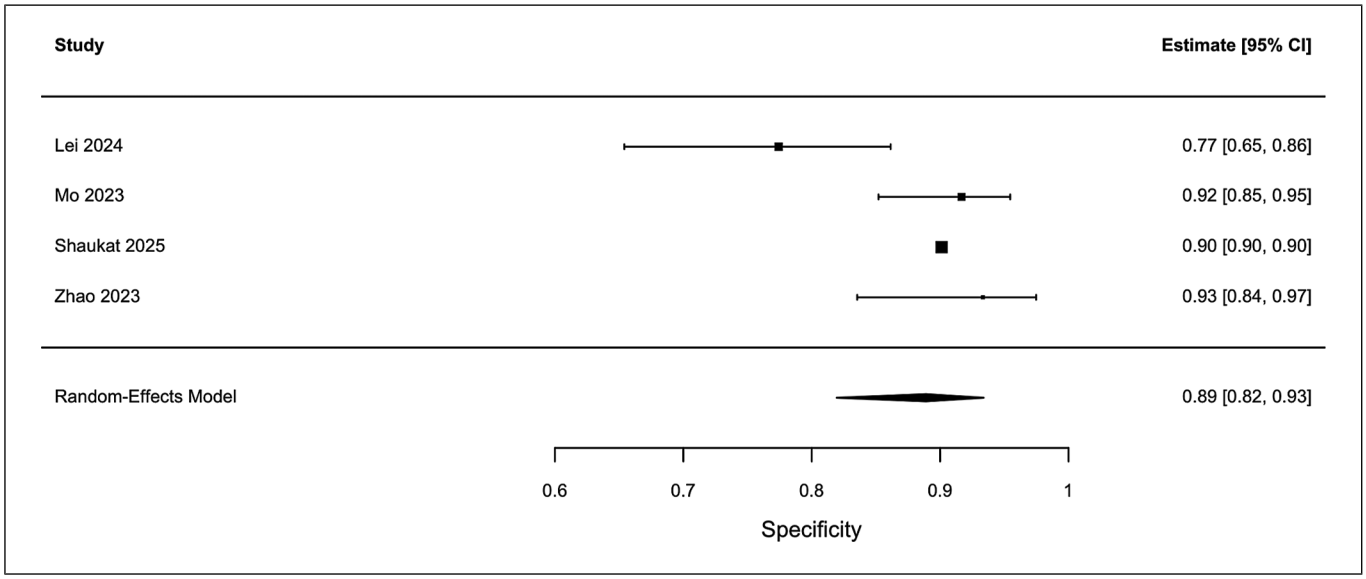


Fig. 5. Forest plot of pooled specificity for methylation-based ctDNA assays (group A). Individual study estimates with 95% confidence intervals are shown; the diamond represents the pooled random-effects estimate.

whether these newer platforms reliably limit FPs. Figure 9 shows the pooled specificities for fragmentomic ctDNA assays.

In Figure 9, we present the specificities found within two different fragmentomic studies, namely, 0.90 (95% CI, 0.80–0.96) and 0.92 (95% CI, 0.88–0.95). Both belong to a narrow range of high specificity, supported by overlapping confidence intervals, indicating agreement between the studies. The pooled value was 0.90 (95% CI, 0.86–0.93), demonstrating a consistently high ability to exclude individuals who did not have cancer. The diamond is centered around 0.90 and is shaped with a

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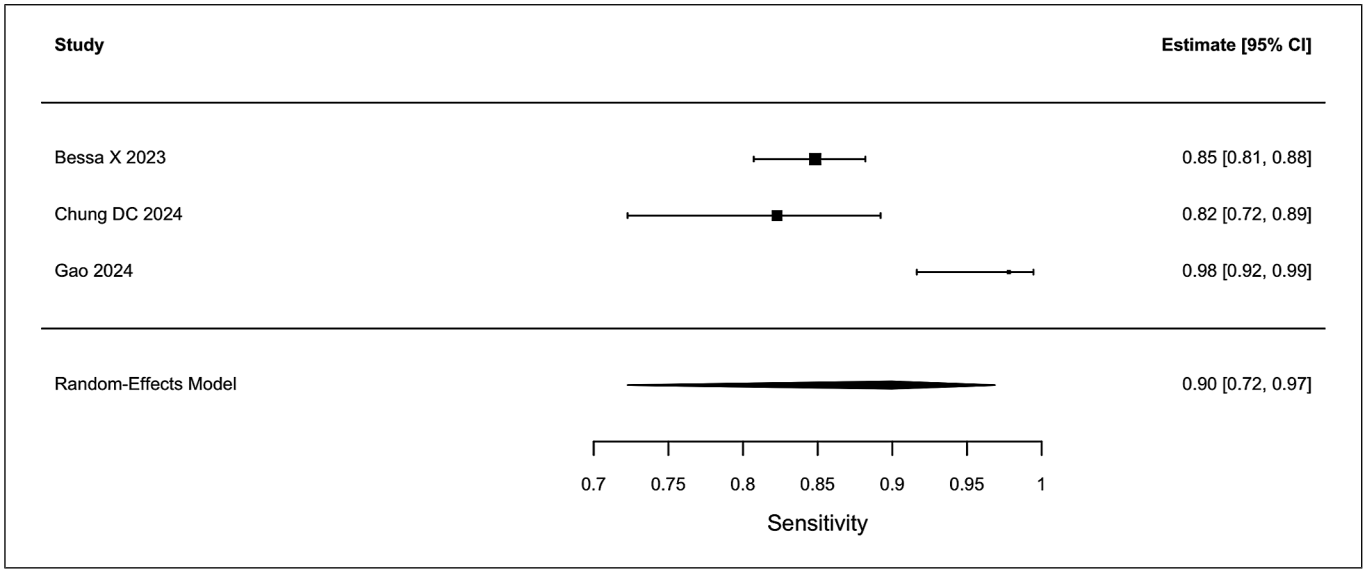


Fig. 6. Forest plot of pooled sensitivity for multimodal ctDNA assays (group B). Study-level sensitivity estimates with 95% confidence intervals are shown; the diamond represents the pooled random-effects estimate.

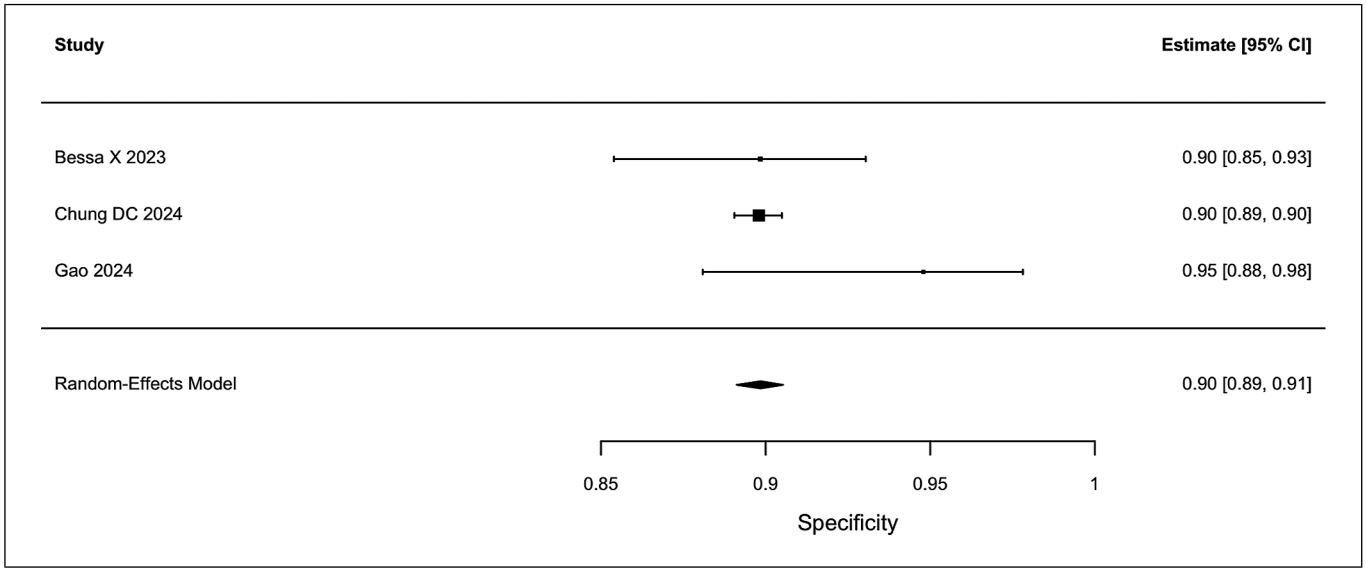


Fig. 7. Forest plot of pooled specificity for multimodal ctDNA assays (group B). Study-level specificity estimates with 95% confidence intervals are shown; the diamond represents the pooled random-effects estimate.

narrow base, illustrating precision and low heterogeneity. Taken together with the pooled sensitivity found in Figure 9, a fragmentomic ctDNA assay has relatively intact performance, indicating a reasonable balance between a high detection capability while preserving reasonable or reliable specificity.

To better identify potential ambiguity stemming from heterogeneity, we stratified studies by assay type and conducted a subgroup analysis. This subgroup analysis created an opportunity to compare methylation-based, multimodal, and fragmentomic-based approaches, in addition to providing pooled

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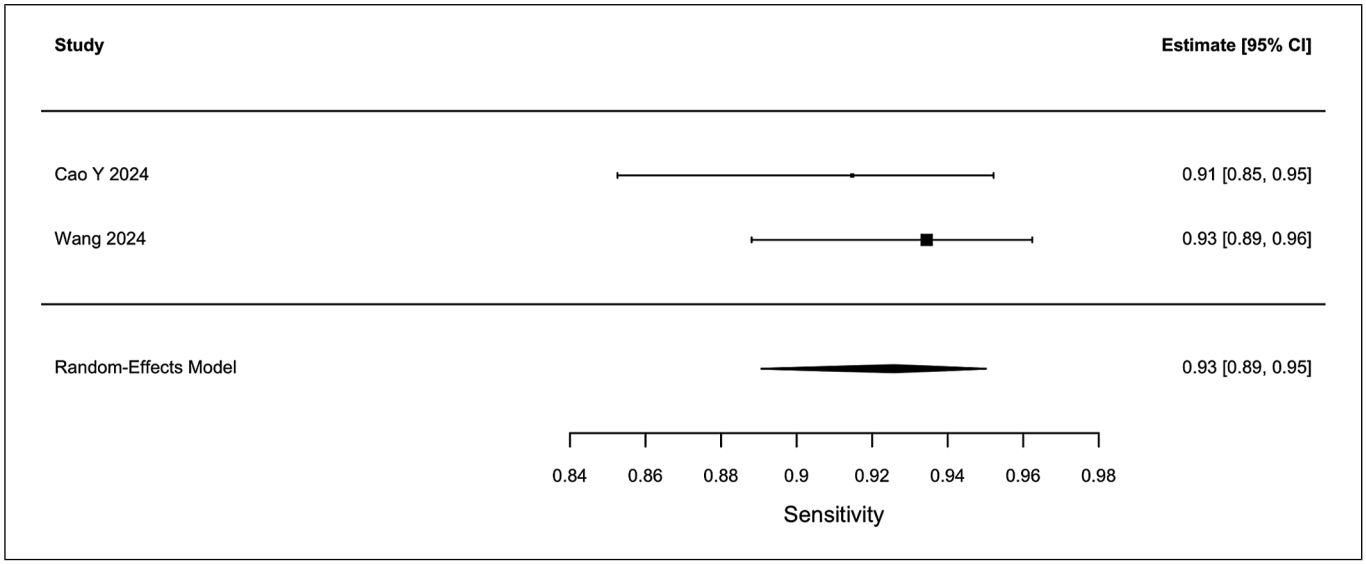


Fig. 8. Forest plot of pooled sensitivity for fragmentomic ctDNA assays (group C). Individual study estimates with 95% confidence intervals are shown; the diamond represents the pooled random-effects estimate.

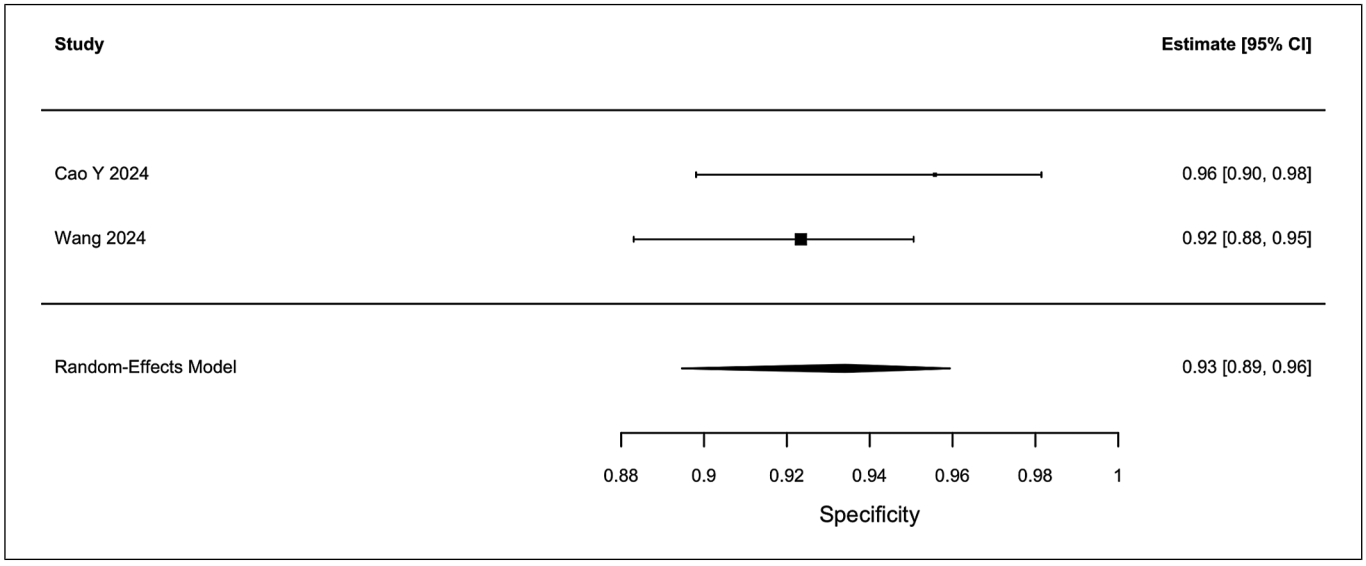


Fig. 9. Forest plot of pooled specificity for fragmentomic ctDNA assays (group C). Study-level specificity estimates with 95% confidence intervals are shown; the diamond represents the pooled random-effects estimate.

sensitivity and specificity for the different approaches. Online supplementary Table S2 summarizes these data. The bivariate Reitsma model integrates sensitivity and specificity into a joint framework while adjusting for study-level variance. This model provides robust pooled estimates together with variance parameters that reflect heterogeneity. Online supplementary Table S3 summarizes the pooled diagnostic accuracy obtained from this model.

Meta-Regression

Meta-regression with assay type (Reitsma) did not find statistically significant differences in sensitivity or false-positive rate by assay type (assay_type B and C coefficients nonsignificant; $p > 0.1$); though subgroup point estimates suggest numerically higher sensitivity/specificity for fragmentomics (C), they were not statistically significant given current data and small subgroup

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sizes (C = 2 studies only.) Online supplementary Table S4 summarizes the regression coefficients, standard errors, and confidence intervals for this analysis.

Publication Bias and Sensitivity Analyses

To assess potential publication bias and small-study effects, funnel plots were constructed for all pooled diagnostic outcomes, including sensitivity, specificity, DOR, PLR, NLR, and AUC (online suppl. Figures S7, S8, S9, S10, S11, S12). Visual inspection did not indicate strong asymmetry for DOR or PLR, suggesting minimal small-study bias, whereas PPV/NPV plots showed mild asymmetry consistent with their high heterogeneity and prevalence dependence. Egger's test was not performed because fewer than ten studies contributed to most pooled outcomes, limiting its statistical power and interpretive reliability.

Sensitivity Analysis (Risk of Bias)

To validate the robustness of the results against potential patient selection bias, we performed a sensitivity analysis excluding studies classified as "high risk" in the patient selection domain of QUADAS-2 (excluding $n = 5$ studies with case-control or enriched designs). The analysis of the remaining studies ($n = 4$; Chung et al. [21], Mo et al. [24], Shaukat et al. [25], Zhao et al. [27]) yielded a pooled sensitivity of 85.7% (95% CI 79.4%–90.4%) and a pooled specificity of 90.0% (95% CI 89.7%–90.4%). These results are consistent with the primary analysis (sensitivity 88.7%; specificity 91.3%), suggesting that the inclusion of case-enriched studies did not substantially inflate the diagnostic performance estimates.

Predictive Values at Screening Prevalence

For CRC screening prevalence of 0.5%, 1%, and 2%, PPV remained below 20%, while NPV consistently exceeded 99%. This suggests that ctDNA assays may serve as robust rule-out tests but are limited as rule-in tools at population prevalence levels. To illustrate the practical implications of test performance at real-world screening prevalence levels, we estimated pooled predictive values. This analysis clarifies how ctDNA assays perform when applied to low-prevalence populations. Online supplementary Table S5 summarizes the PPV and NPV at assumed prevalence rates of 0.5%, 1%, and 2%.

Certainty of Evidence

For each pooled diagnostic accuracy result, we rated the certainty of evidence using the GRADE approach. The certainty was then judged from moderate (pooled sensitivity, specificity, and NPV) to low (DOR, PLR, NLR, and PPV), principally due to heterogeneity and inconsistency dissociation as well as inaccuracy of the estimates. The results are summarized in the most detail in online supplementary Table S6.

We used the GRADE approach to assess the certainty of evidence of pooled diagnostic estimates. The confidence of the overall evidence differed according to outcomes, due mostly to heterogeneity, inconsistency and imprecision across studies.

Summary

Blood-based ctDNA assays show high pooled sensitivity (~89%) and specificity (~91%) for detection of CRC in the included asymptomatic/early-detection cohorts, with overall strong discrimination (DOR ≈ 84; AUC ≈ 0.97). Likelihood ratios (PLR ≈ 9.2; NLR ≈ 0.12) indicate that ctDNA testing in these studies substantially changes posttest probability. However evidence was assessed as moderate for sensitivity, specificity, and NPV, and judged low for the other outcomes, which also reflect inconsistency and imprecision, but with substantial between-study heterogeneity (especially for PPV/NPV and DOR) owing to assay platform, thresholds, populations, small numbers in strata, etc.

Overall risk of bias across studies was judged moderate. The most frequent concerns arose in the patient selection domain, where several cohorts were enriched with known CRC cases and healthy controls, limiting applicability to true screening populations. Some studies also lacked blinding of index test interpretation, introducing potential index test bias. The reference standard (colonoscopy or histopathology) was generally appropriate, but reporting of flow and timing was often incomplete. Detailed domain-level judgments for each study are summarized in online supplementary Table S7.

Stage-Specific Sensitivity

A subset of studies reported ctDNA assay sensitivity stratified by tumor stage. Table 3 summarizes stage-specific sensitivity values where available. Several studies provided only combined estimates for early stages (I–II) or reported aggregated stages (I–III); these were recorded accordingly. Because not all studies reported per-stage data, results are presented descriptively in Table 3.

Stage-specific sensitivity values of ctDNA assays, summarized in Table 3, were further visualized to highlight trends across tumor stages (online suppl. Fig. S13). Although not all studies reported per-stage estimates, the pooled data provide useful insights

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Table 3. Stage-specific sensitivity of ctDNA assays (studies reporting stages I–IV where available)

Study (author)	Assay type	Stage I sensitivity	Stage II sensitivity	Stage III sensitivity	Stage IV sensitivity	Notes
Bessa et al. [19]	Multimodal	84% (41/49)	94% (184/196)	96% (70/73)	NR	Stage IV too few cases (n = 2), not reported
Shaukat et al. [25]	Methylation	NR	NR	NR	NR	Only overall sensitivity reported, no stage breakdown
Zhao et al. [27]	Methylation	NR	NR	NR	NR	Early-stage (I+II) combined sensitivity given, no per-stage detail
Mo et al. [24]	Methylation	85.1%	79.3%	91.8%	NR	Stage IV not reported
Gao et al. [22]	Multimodal	NR	89.2%	NR	NR	Only stage II sensitivity reported clearly
Cao et al. [20]	Fragmentomic	NR	NR	NR	NR	No per-stage sensitivity available
Øgaard et al. [28]	Mixed	NR	NR	NR	NR	No clear per-stage sensitivity reported
Wang et al. [26]	Fragmentomic	~35%	NR	46.1%	NR	Reported in context of comparison, needs cautious interpretation
Lei et al. [23]	Methylation	NR	NR	NR	NR	No stage-specific values provided
Chung et al. [21]	Multimodal	55%	NR	NR	100% (10/10)	Stage II/III not reported

Values reflect study-level percentages and sample counts (n), with notes provided where applicable.

into sensitivity patterns across disease progression [29]. The combined findings in online supplementary Figure S13 clearly demonstrate that as tumor stage increases, so does the sensitivity of the ctDNA assay.

Because it can be difficult to detect little tumor shedding at very early stages of illness, detection is low in stage I (≈65%). It is important to note that this estimate is derived only from the subset of studies that provided stage-specific stratification, as several large included studies did not report granular data for early-stage disease. When the tumor burden is at its maximum, in Stage IV (100%), sensitivity approaches near-perfect accuracy after improving significantly in stage II (≈88%) and stage III (≈78%). These patterns illustrate the limitations of early-stage screening while also reinforcing the biological gradient of ctDNA release and showing its potential for monitoring advanced disease. Crucially, these estimates should be interpreted cautiously because variation in assay platforms may contribute to variability and not all studies provided complete per-stage data (see Table 3). Nevertheless, the general pattern offers a strong indication that ctDNA sensitivity increases as tumor size increases.

Discussion

Principal Findings

Blood assay testing for ctDNA in this meta-analysis for the early detection of CRC showed a very high overall pooled diagnostic accuracy with the sensitivity of 0.89 and the specificity of 0.91 and a great overall discrimination (AUC about 0.97). The combined PLR (about 9.2) and low NLR (approximately 0.12) show that there are clinically relevant probability changes after a positive or negative result. However, significant between-study heterogeneity was observed for DOR and predictive values (PPV/NPV), and subgroup analyses did not reveal significant differences across assay types.

Importantly, several included studies were conducted in diagnostic or high-risk cohorts rather than true average-risk screening populations. This distinction is clinically relevant, as diagnostic accuracy observed in enriched cohorts may overestimate performance when applied to population-based screening settings. Accordingly, pooled estimates should be interpreted as reflecting test performance under study conditions rather than definitive screening accuracy.

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Sensitivity for stage I CRC was consistently lower than for more advanced stages, reflecting limited tumor DNA shedding in early disease. This represents a biological constraint of ctDNA-based detection rather than a technical shortcoming of individual assays. These findings highlight the challenge of detecting very early tumors and underscore the need for cautious interpretation of early-stage screening performance.

Although NPV remained high across screening prevalence scenarios, PPV was modest at population prevalence levels. This implies that widespread ctDNA screening could result in a substantial number of follow-up colonoscopies per cancer detected, with important implications for healthcare resource utilization and patient burden.

When compared with the FIT which typically shows sensitivities between 70% and 79% depending on threshold and population, pooled ctDNA sensitivity appears favorable, suggesting its potential as a promising alternative. Nonetheless, direct head-to-head comparisons within identical screening cohorts are lacking and are essential before definitive conclusions can be drawn. At present, FIT remains the more cost-effective and accessible tool, while ctDNA assays are more expensive and require specialized laboratory infrastructure and standardization [30]. Furthermore, given the low prevalence of CRC in screening populations, the PPV of ctDNA testing remains modest (e.g., ≈5% at 0.5% disease prevalence), resulting in more FPs and an added colonoscopy burden at the population level.

Included studies differed substantially in design, targeted biomarkers (mutation, methylation, fragmentomics), and analytical pipelines [31]. Although multimodal and fragmentomic approaches showed slightly higher pooled accuracy, subgroup sizes were limited, and meta-regression did not demonstrate statistically significant differences. Tumor stage strongly influenced ctDNA detection rates, with sensitivity increasing alongside tumor burden. However, incomplete reporting of stage-specific data across studies limited precise estimation of early-stage (stage I–II) performance, which is most critical for effective screening.

Implementing ctDNA testing in real-world screening programs would require generous investment in laboratory capacity, trained personnel, and consistent quality assurance. While multimodal strategies, such as combining ctDNA with FIT or clinical risk models, may help balance sensitivity, specificity, and downstream colonoscopy burden, their cost-effectiveness and operational feasibility have yet to be validated through large-scale screening trials [32].

Strengths and Limitations

This review has several methodological strengths: a prespecified bivariate model for paired pooling of sensitivity and specificity, appropriate transformations and zero-cell corrections, inclusion of both bivariate and inverse-variance pooled metrics, and sensitivity analyses through subgroup and meta-regression approaches. High risk of bias in patient selection, particularly from case-control designs, may overestimate diagnostic performance compared with true screening settings. Additionally, incomplete reporting of stage I sensitivity in several key studies represents a major evidence gap for informing screening guidelines. We also projected PPV and NPV at plausible population-level screening prevalences, enhancing clinical interpretability.

However, moderate-to-high heterogeneity across studies, small subgroup sample sizes (particularly for fragmentomics), and incomplete reporting of certain outcomes limited the precision of pooled estimates. Most pooled outcomes included fewer than ten studies, restricting power for small-study bias assessment (e.g., Egger's test).

Several important limitations should be acknowledged. First, many included studies were conducted in case-enriched or high-risk cohorts rather than true average-risk screening populations. This may inflate diagnostic accuracy estimates and limits direct generalizability to population-based screening programs. Second, a substantial proportion of included studies originated from East Asia, particularly China, which may restrict applicability across different healthcare systems and ethnic populations. Third, studies without full-text availability were excluded, introducing potential selection bias; however, abstracts alone rarely provide sufficient methodological and diagnostic detail for reliable pooling. Finally, stage-specific data especially for stage I disease and advanced adenomas were inconsistently reported, limiting robust conclusions regarding very early cancer detection.

Future Directions

Future research should focus on large, prospective, population-based screening trials that accurately estimate test performance in asymptomatic individuals. The ctDNA assays' standardization which encompasses biomarker panels, thresholds, specimen types, and reporting formats will be a major supporting factor for comparability and pooling of

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studies [28]. The stage-specific analyses, particularly for stage I cancers and precancerous lesions, are needed to clarify the actual early-detection value. The combination of ctDNA with the currently used techniques like FIT, colonoscopy, or risk algorithms could be a way to increase sensitivity and effectively use the resources. Finally, comprehensive cost-effectiveness studies and technological advances in ultra-sensitive, multimodal, and methylation-based assays will determine whether ctDNA testing can be sustainably integrated into population-level screening frameworks [33].

Conclusion

The use of blood-based ctDNA assays for CRC detection has been shown to be remarkably accurate diagnostically, with a pooled sensitivity of baseline around 89%, a specificity of 91%, and an AUC of 0.97. The related PLR and NLR indicate sizable shifts in posttest probability that are clinically relevant, thus proving the potential of ctDNA as a noninvasive screening tool that might be accepted in the future. But there are still some restrictions to their routine use as primary screening strategy due to considerable variability among studies, lack of data for some assay subtypes, the nature of predictive values which is dependent on prevalence, and the current lack of large-scale population-based validations. Future research will need to focus on setting up standardized assay protocols, conducting direct comparisons with FIT in population cohorts, assessing stage-specific performance especially for stage I disease cost-effectiveness analyses, and undertaking large prospective validation studies to establish the actual clinical utility of ctDNA testing in CRC screening.

Statement of Ethics

The present study sourced all data from publicly available and free-to-use sources. Thus, the study was deemed exempt from ethical approval. All authors consent to publication and abide by the terms and conditions as specified by the journal. The authors are accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

Conflict of Interest Statement

The authors declare no competing interests.

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Author Contributions

Y.G. and R.N. conceived the study, led its design, coordinated all research activities, and together with V.P., S.K.M., T.M., A.Y., M.C., M.K., O.O., and M.H. performed material preparation, data collection, and initial analyses. The first draft was written by Y.G., R.N., V.P., S.K.M., T.M., A.Y., M.C., M.K., O.O., and M.H. All authors reviewed all the versions of this manuscript and have read and approved the final manuscript.

Data Availability Statement

All data generated or analyzed during this study are included in this article and its supplementary material files. Further inquiries can be directed to the corresponding author.

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